# ESWCap combined C+R front end (STM32G474 carrier)

**REV C: USB-only; 4-wire C/R; OPAMP3 SH buffer; J5 FH/SH/SL/FL; PA0=DRIVE, PA6=BIAS_MON**

C: DRIVE -> C mux -> Rsense -> Vcap -> C1/C2 -> K1B -> N_DUT -> DUT -> B -> K1A -> GND

**REV B: C1 = 1000 uF / 35 V BIPOLAR (+ marking not used).**

D6 (1N4148) blocks Q1 body diode; R9 = 47R 2W; C3 returns to VREF.

Console J7: PA9=TX, PA10=RX, GND, +3V3. See docs/pcb-design.md.

R: +3V3 -> Q1 -> N_DUT -> DUT -> B -> K2A -> R mux -> RSENSE -> GND. OPAMP1: VINP=B(PA7), VINM=VREF(PA3), VOUT=PA2. TIA: LMC6482 A.

J2
1 GND
2 +3V3
3
4
5
6 DRIVE
7 VCAP_ADC
8 OP1_OUT
9 VREF
10 BIAS_SET
11 VBIAS
12 TIA_OUT
13 MUXR_COM
14 C_MODE_SW
15 VN_SENSE
16
17 VBUF
18
19 K2_DRV
20 SSR13
WeAct P2

J1
1 GND
2 K1_DRV
3 MUXR_A1
4 MUXR_A0
5 MUXC_EN
6 MUXC_A1
7 MUXC_A0
8 BIAS_UART_RX
9 BIAS_UART_TX
10 BIAS_EN
11
12
13 UART_RX
14 UART_TX
15 BIAS_MON
16
17 MUXR_EN
18 R_FORCE
19 SSR2_CTRL
20 VBUF
WeAct P1

J3
1 GND
2 GND
3
4
WeAct P3 (NC)

J4
1 GND
2 GND
3 +3V3
4 +3V3
WeAct P4 3V3

J7
1 UART_TX
2 UART_RX
3 GND
4 +3V3
Console

U2
74LVC1G34
+3V3
DRIVE 2
4 C_MUX_COM
3 GND
5
1

U3
MAX4619
C_CH3 1 Y1
C_CH2 2 Y0
MUXC_Q 3 Z1
C_MUX_COM 4 Z
MUXC_P 5 Z0
MUXC_EN 6 ENABLE
7 NC
GND 8 GND
VCC 16 +3V3
Y 15 MUXC_Q
X 14 MUXC_P
X1 13 C_CH1
X0 12 C_CH0
A 11 MUXC_A0
B 10 MUXC_A0
C 9 MUXC_A1

U4
MAX4619
R_CH3 1 Y1
R_CH2 2 Y0
MUXR_Q 3 Z1
MUXR_COM 4 Z
MUXR_P 5 Z0
MUXR_EN 6 ENABLE
7 NC
GND 8 GND
VCC 16 +3V3
Y 15 MUXR_Q
X 14 MUXR_P
X1 13 R_CH1
X0 12 R_CH0
A 11 MUXR_A0
B 10 MUXR_A0
C 9 MUXR_A1

U5
AQY212
SSR1_LED 1 LED+
GND 2 LED-
SW 3 SSR1_SW
SW 4 V_BIAS

U6
AQY212
SSR3_LED 1 LED+
GND 2 LED-
SW 3 SSR3_SW
SW 4 VCAP

U7
AQY212
SSR2_LED 1 LED+
GND 2 LED-
SW 3 SSR2_SW
SW 4 FH

U9
AO3401
Q1_D 1 +3V3
2 U9_LED_K
LED+ 11 SW 3 SH
LED- SW 4 VN_IN

U10
AQY212
1 +3V3 LED+
2 U10_LED_K LED-
SW 3 SH_DIV
SW 4 VN_SENSE

K1
NC1 3 GND
COM1 4 FL
NO1 5 C_BLOCK
NC2 10 FH
COM2 9
NO2 8
COIL+ +3V3
COIL- 12 K1_C

K2
NC1 3
COM1 4 FL
NO1 5 MUXR_COM
NC2 10 RF_TOP
COM2 9 TIA_OUT
NO2 8
COIL+ +3V3
COIL- 12 K2_C
KELVIN

U1A
LMC6482
IN+ 3 VBIAS
IN- 2 FL
OUT 1 TIA_OUT

U1B
LMC6482
IN+ 5 VBUF
IN- 6
OUT 7 VBUF

U1C
LMC6482
V+ 8 +3V3
V- 4 GND

Q1
BC847
G Q1_GATE
S +3V3
D Q1_D

Q2
BC847
B K1_BASE
C K1_C
E GND

Q3
BC847
B K2_BASE
C K2_C
E GND

D1
BAV199
A1 GND
K2 +3V3
COM VCAP

D2
BAV199
A1 GND
K2 +3V3
COM VN_SENSE

D3
BAT54S
A1 GND
K2 +3V3
COM TIA_OUT

D4
1N4148
K1_C +3V3

D5
1N4148
K2_C +3V3

D6
1N4148
VCAP C_BLOCK
FH

J5
DUT
1 FH
2 SH
3 SL
4 FL

J6
V_BIAS 0–20V
1 V_BIAS
2 GND

J8
Pin_1 1 BIAS_SET_F
Pin_2 2 BIAS_EN
Pin_3 3 BIAS_MON_F
Pin_4 4 GND
Pin_5 5 +3V3
Pin_6 6 BIAS_UART_TX
Pin_7 7 BIAS_UART_RX
Pin_8 8 GND

R1 C_CH2 1 2 VCAP 98.9 1%
R2 C_CH1 1 2 VCAP 993.6 1%
R3 C_CH0 1 2 VCAP 98.6k 1%
R4 C_CH3 1 2 VCAP 1.01M 1%
R5 R_CH0 1 2 GND 98.9 0.1%
R6 R_CH1 1 2 GND 993.6 0.1%
R7 R_CH2 1 2 GND 98.6k 0.1%
R8 R_CH3 1 2 GND 4.7M 0.1%
R9 SSR1_SW 1 2 FH 47
R10 SSR2_SW 1 2 GND 100
R11 SSR3_SW 1 2 GND 1
R12 V_BIAS 1 2 FH 4.8M
R13 +3V3 1 2 Q1_GATE 100k
R14 OP1_OUT 1 2 Q1_GATE Q1_GATE 1k
R15 R_FORCE 1 2 Q1_GATE 10k
R16 FL 1 2 RF_TOP 10M 0.1%
R17 +3V3 1 2 VREF 7.15k
R18 VREF 1 2 GND 1k
R19 +3V3 1 2 VBIAS 10k
R20 VN_IN 1 2 VN_SENSE 1k
R21 VBIAS 1 2 GND 10k
R22 SSR13 1 2 SSR1_LED 330
R23 SSR13 1 2 SSR3_LED 330
R24 SSR2_CTRL 1 2 SSR2_LED 330
R25 K1_DRV 1 2 K1_BASE 1k
R26 K2_DRV 1 2 K2_BASE 1k
R27 K1_BASE 1 2 GND 10k
R28 K2_BASE 1 2 GND 10k
R29 GND 1 2 R_MODE 10k
R30 R_MODE 1 2 +3V3 100k
R31 +3V3 1 2 VCAP 100k
R32 VCAP 1 2 VCAP_ADC 1k
R33 VCAP 1 2 C_BLOCK 10M 0.1%
R34 BIAS_SET BIAS_SET_F 1k
R35 2 GND 1 BIAS_EN 10k
R36 BIAS_MON BIAS_MON_F 1k
R38 SH 2 SH_DIV 10M 0.1%
R39 SH_DIV 2 GND 1M
R43 C_MODE_SW U10_LED_K 330

C1 VCAP 1 2 C_BLOCK 1000uF/35V BP
C2 VCAP 1 2 C_BLOCK 68nF
C3 OP1_OUT 1 2 VREF 100pF
C4 FL 1 2 RF_TOP 10pF
C5 VREF 1 2 GND 100nF
C6 VBIAS 1 2 GND 100nF
C7 VN_SENSE 1 2 GND 1nF
C8 +3V3 1 2 GND 100nF
C9 +3V3 1 2 GND 10uF
C10 +3V3 2 GND 100nF
C11 +3V3 1 2 GND 100nF
C12 +3V3 1 2 GND 100nF
C13 +3V3 1 2 GND 100nF
C14 +3V3 1 2 GND 100nF
C15 +3V3 2 GND DRIVE 10uF
C16 BIAS_SET BIAS_SET_F 1k
C17 BIAS_MON BIAS_MON_F 10nF
C20 SH_DIV 2 GND 100pF
C21 SH 2 SH_DIV 10pF

FLG1 +3V3
FLG3 GND
FLG4 V_BIAS

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File: Project.kicad_sch
Title: ESWCap C+R Meter — carrier board
Size: A3 | Date: 2026-11-14 | Rev: C.1
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